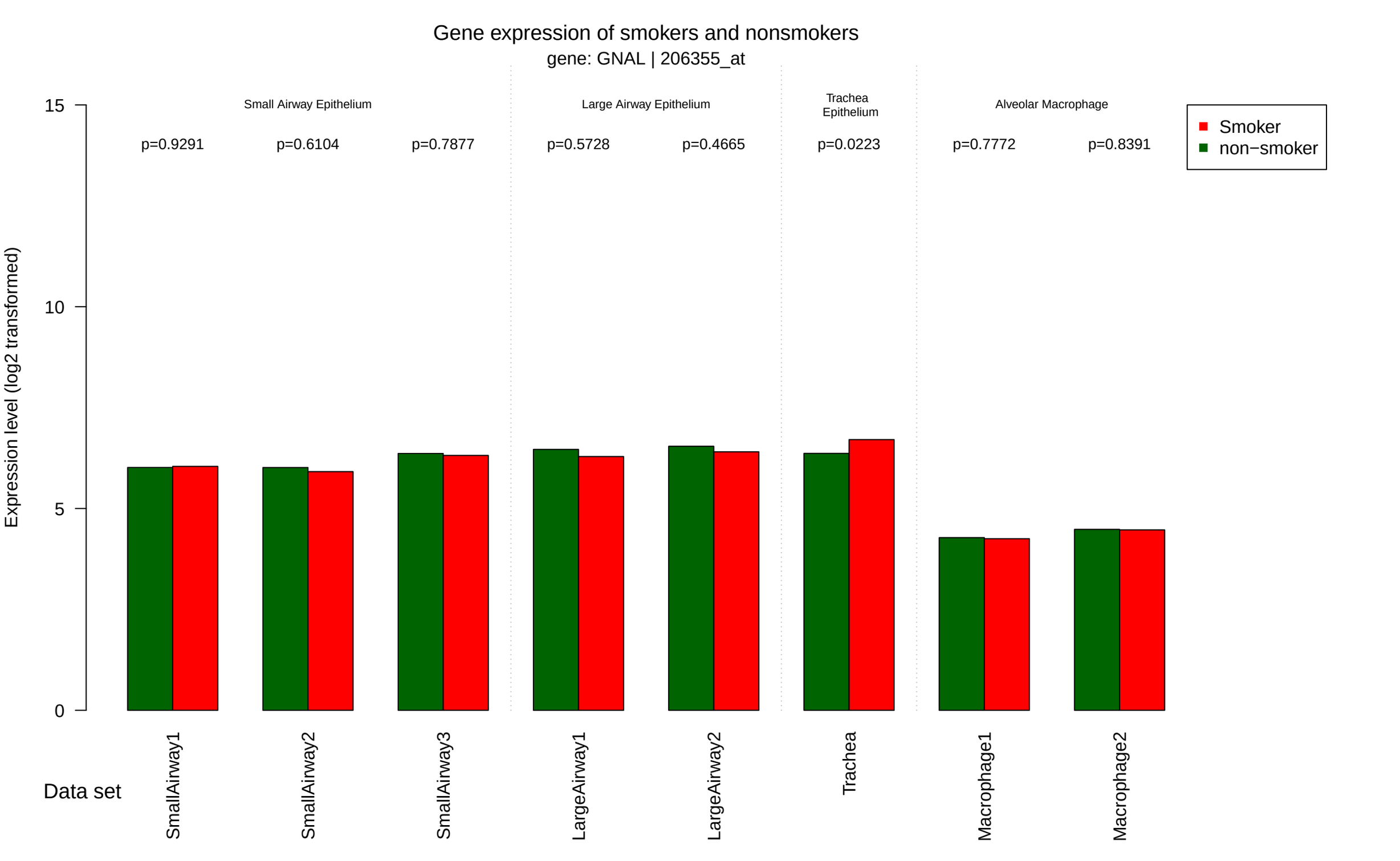
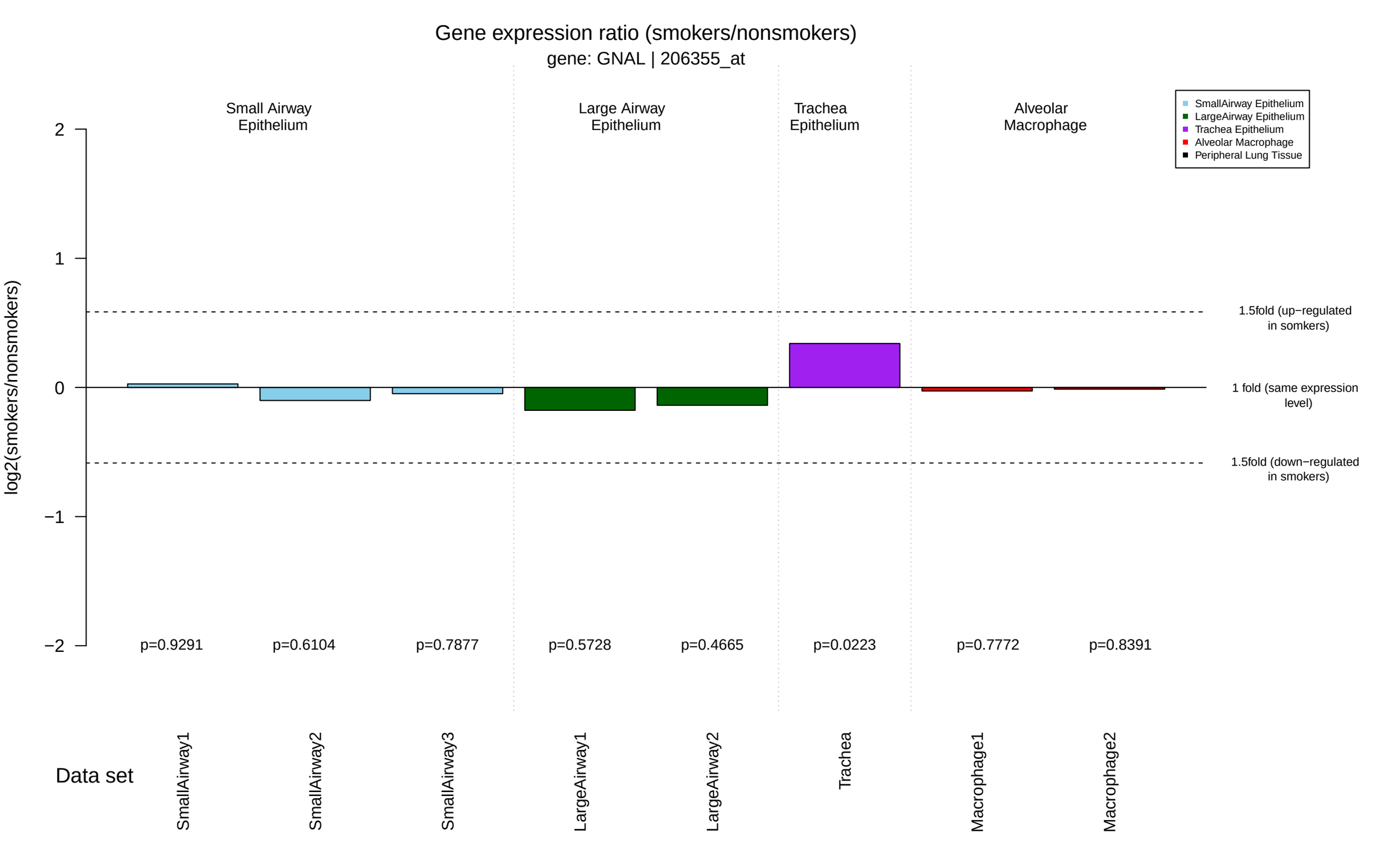
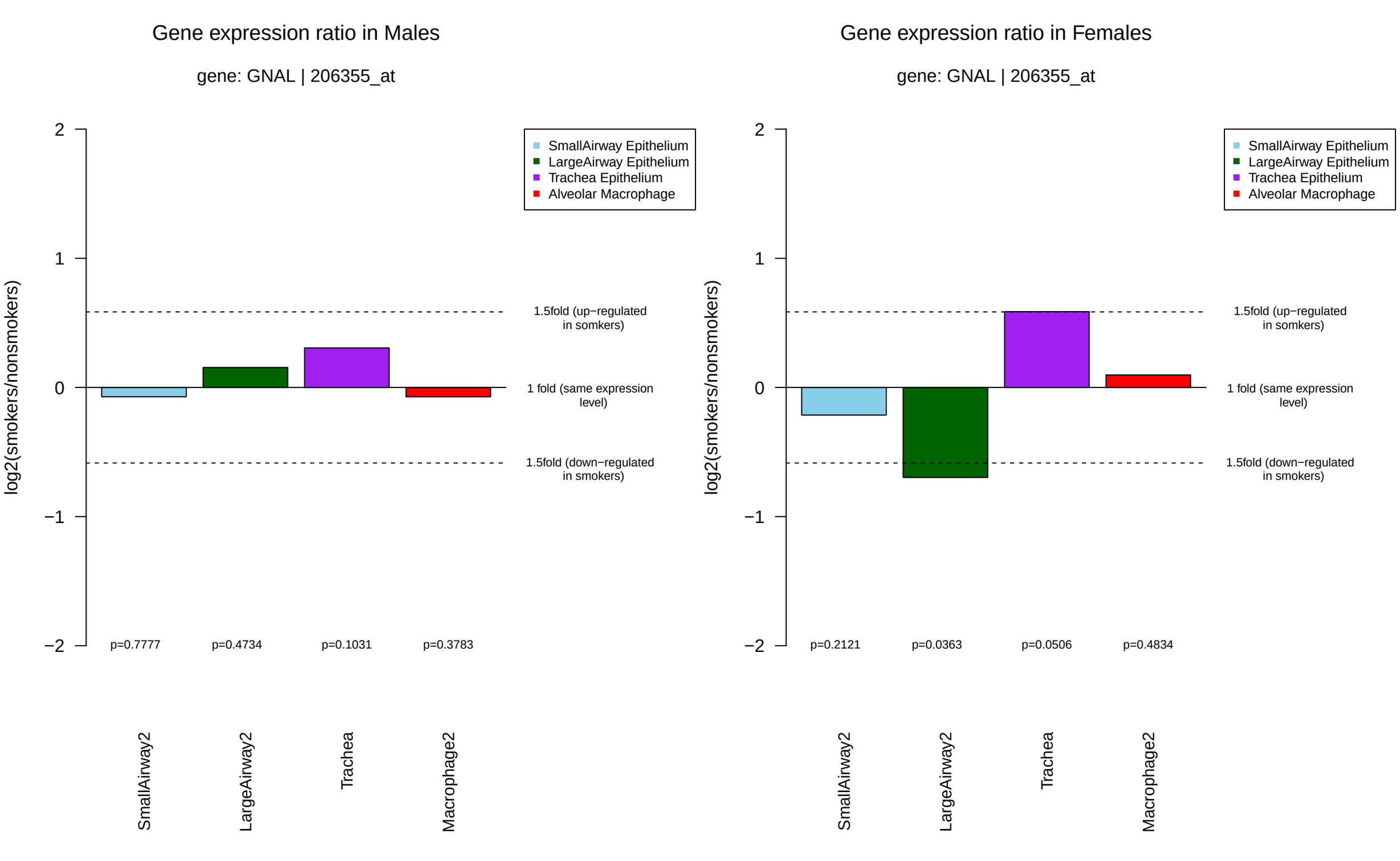
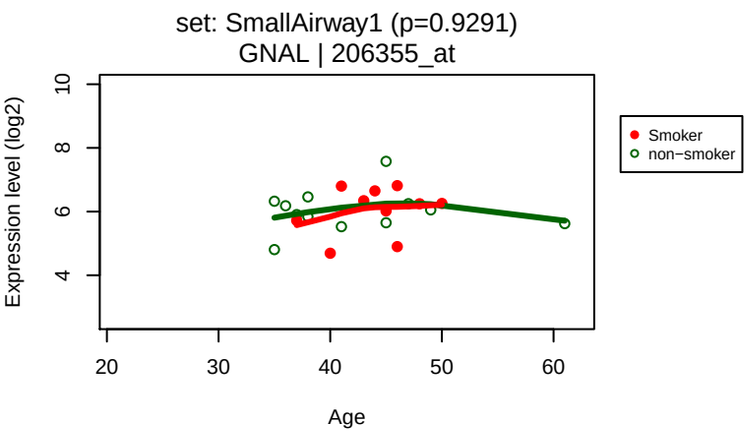
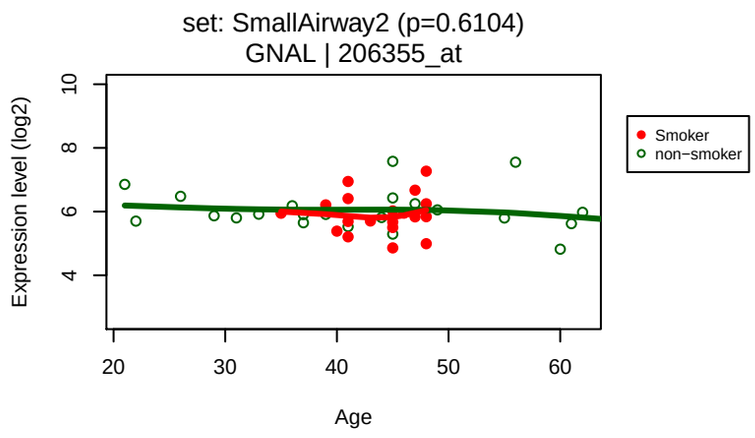
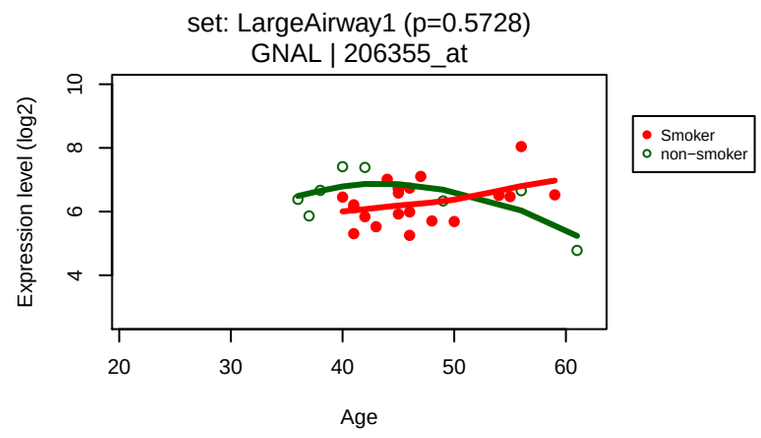

Gene expression of smokers and nonsmokers
gene: GNAL | 206355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9291
p=0.6104
p=0.7877
p=0.5728
p=0.4665
p=0.0223
p=0.7772
p=0.8391
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

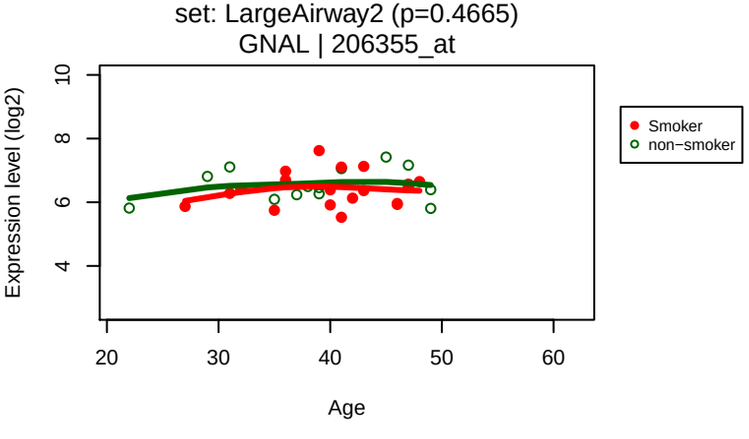

Gene expression ratio (smokers/nonsmokers)
gene: GNAL | 206355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9291
p=0.6104
p=0.7877
p=0.5728
p=0.4665
p=0.0223
p=0.7772
p=0.8391
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

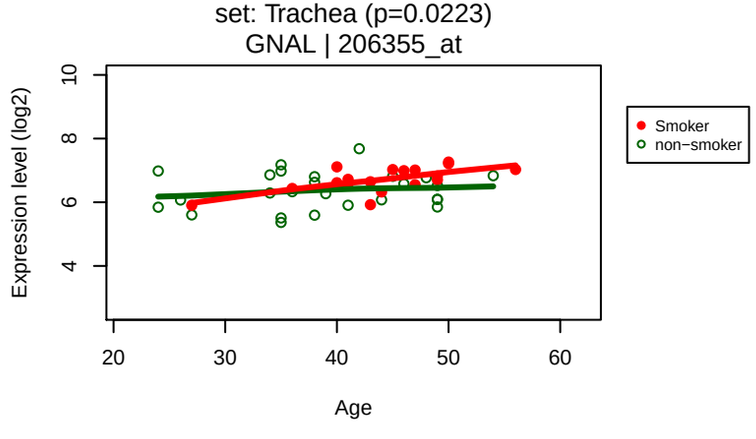

Gene expression ratio in Males
gene: GNAL | 206355_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7777
p=0.4734
p=0.1031
p=0.3783
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAL | 206355_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2121
p=0.0363
p=0.0506
p=0.4834
SmallAirway2
LargeAirway2
Trachea
Macrophage2

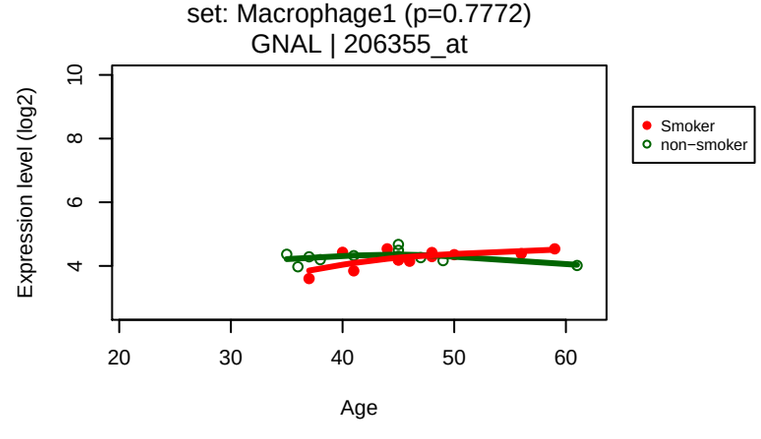

set: SmallAirway1 (p=0.9291)
GNAL | 206355_at
Expression level (log2)
Age
Smoker
non−smoker

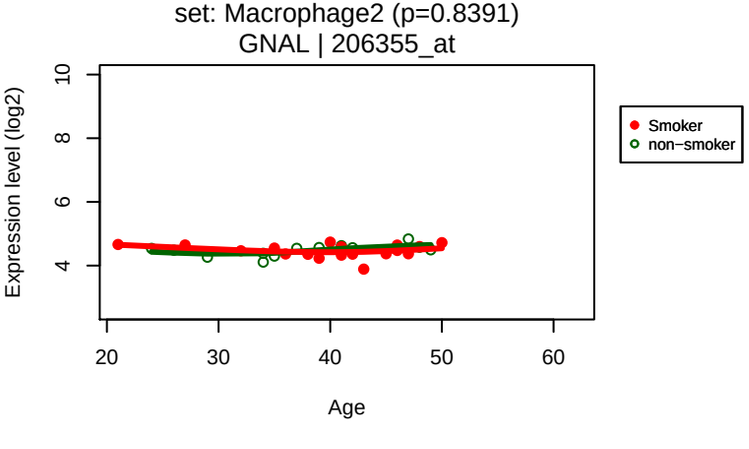

set: SmallAirway2 (p=0.6104)
GNAL | 206355_at
Expression level (log2)
Age
Smoker
non−smoker

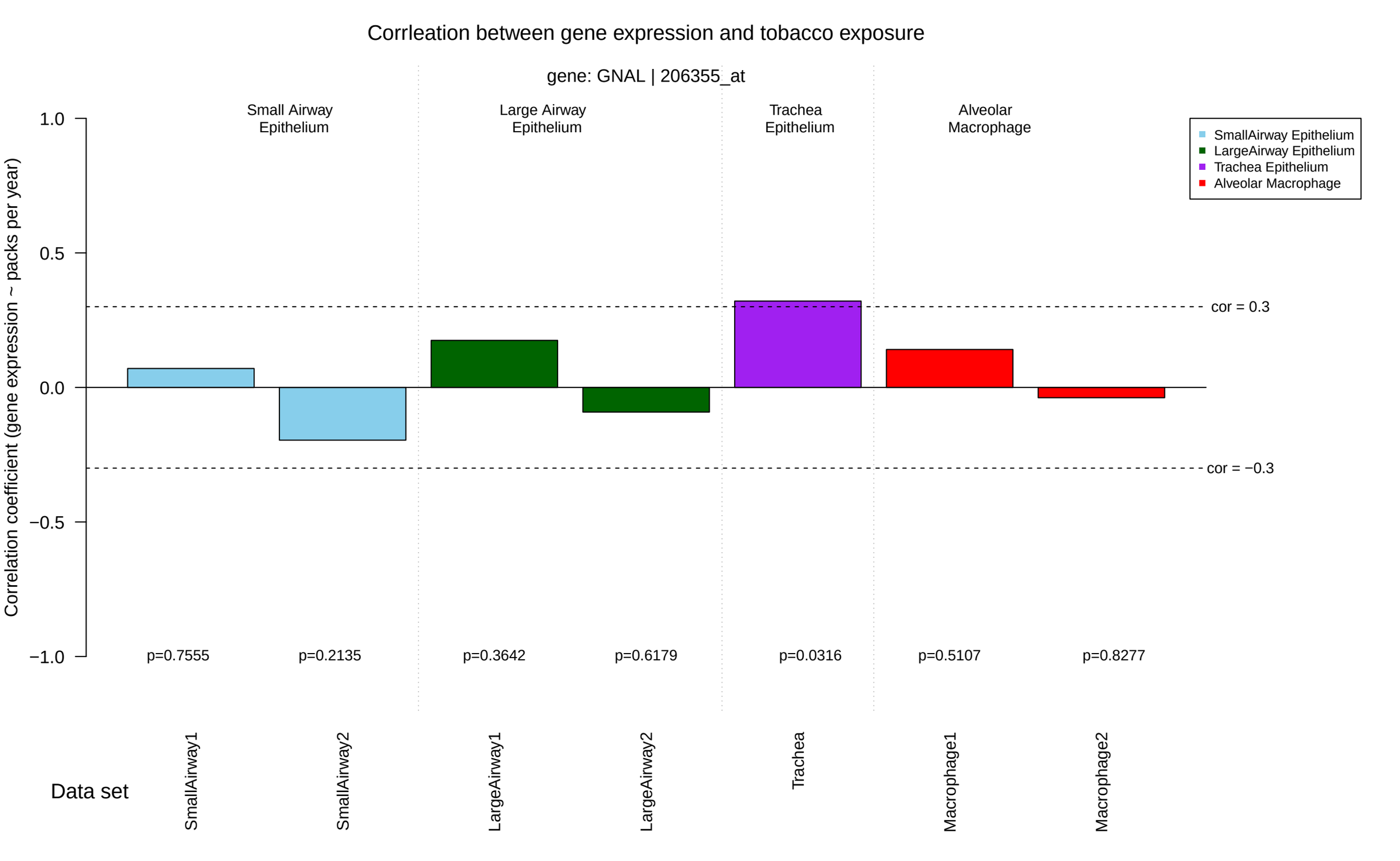

set: LargeAirway1 (p=0.5728)
GNAL | 206355_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4665)
GNAL | 206355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0223)
GNAL | 206355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7772)
GNAL | 206355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8391)
GNAL | 206355_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAL | 206355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7555
p=0.2135
p=0.3642
p=0.6179
p=0.0316
p=0.5107
p=0.8277
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2